

Shunt Capacitance and High Frequency Response

The capacitance of amplifier that provides a path for the AC signal to the ground is called *shunt capacitance*. The shunt capacitances formed at the input and output cause reduction in gain at high frequencies. In Chapter , we have discussed the inter-electrode capacitances of BJT which are mainly due to device structural capacitance. Shunt capacitances are also created by wiring, terminal electrodes, soldering joint, and any other circuit structure where conducting materials come close together. This type of capacitance is called *stray capacitance*.

To study the performance of an amplifier let us represent all input capacitance by C_{in} and all

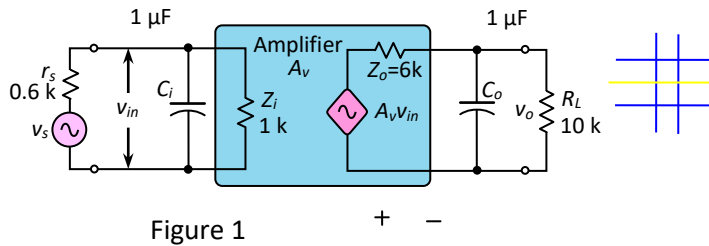


Figure 1

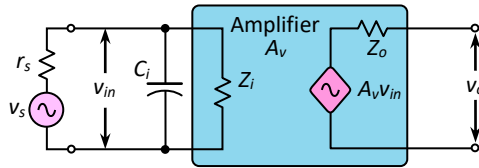


Figure 2

Figure 3

At first we will explain the high frequency response of the amplifier due to C_{in} . As shown in Fig. v_o is open circuit output voltage and $v_o/v_{in} = A_v$. In the mid-band frequency range, reactance of the shunt capacitances are high enough to be neglected and the reactance of the coupling capacitors are low enough to be neglected. At mid-band frequency range, from Fig.10-, we can write,

	$\frac{v_{in}}{v_s} = \frac{Z_i}{r_s + Z_i}$	
	$\frac{v_o}{v_s} = \frac{v_{in}}{v_s} \times \frac{v_o}{v_{in}} = \frac{Z_i A_v}{r_s + Z_i} = A_{v(mid)}$	

At frequencies where the reactance of C_{in} is comparable to Z_i , we have to write

	$\frac{v_{in}}{v_s} = \frac{-jX_{Ci} \parallel Z_i}{r_s - jX_{Ci} \parallel Z_i} = \frac{(-j/\omega C_i) \parallel Z_i}{r_s - (j/\omega C_i) \parallel Z_i}$	
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	$\frac{v_{in}}{v_s} = \frac{-\frac{j}{\omega C_i} Z_i / (Z_i - \frac{j}{\omega C_i})}{r_s - \frac{j}{\omega C_i} Z_i / (Z_i - \frac{j}{\omega C_i})}$	
	$\frac{v_{in}}{v_s} = \frac{-\frac{j}{\omega C_i} Z_i}{r_s \left(Z_i - \frac{j}{\omega C_i} \right) - \frac{j}{\omega C_i} Z_i}$	
	$\frac{v_{in}}{v_s} = \frac{-Z_i}{r_s(Z_i \omega C_i - j) - jZ_i} = \frac{-Z_i}{r_s Z_i \omega C_i - j r_s - j Z_i}$	
	$\frac{v_{in}}{v_s} = \frac{Z_i}{-r_s Z_i \omega C_i + j(r_s + Z_i)}$	

Magnitude of this value is,

	$\left \frac{v_{in}}{v_s} \right = \frac{Z_i}{\sqrt{(\omega r_s Z_i C_i)^2 + (r_s + Z_i)^2}}$	
	$\left \frac{v_{in}}{v_s} \right = \frac{\frac{Z_i}{r_s + Z_i}}{\sqrt{\left(\omega C_i \times \frac{r_s Z_i}{r_s + Z_i} \right)^2 + 1}}$	
	$\left \frac{v_{in}}{v_s} \right = \frac{\frac{Z_i}{r_s + Z_i}}{\sqrt{[\omega C_i (r_s \parallel Z_i)]^2 + 1}}$	

Now,

	$\left \frac{v_o}{v_s} \right = \frac{v_{in}}{v_s} \times \frac{v_o}{v_{in}} = \frac{\frac{Z_i}{r_s + Z_i} A_v}{\sqrt{[\omega C_i (r_s \parallel Z_i)]^2 + 1}} = \frac{A_{v(mid)}}{\sqrt{[\omega C_i (r_s \parallel Z_i)]^2 + 1}}$	
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From this equation, we find that $|v_o/v_s|$ decreases with the increase in frequency. The higher cut-off frequency, ω_h , is the frequency at which the voltage gain falls to $\frac{A_{v(mid)}}{\sqrt{2}}$. Putting this value into above equation

	$\frac{A_{v(mid)}}{\sqrt{[\omega_h C_i (r_s \parallel Z_i)]^2 + 1}} = \frac{A_{v(mid)}}{\sqrt{2}}$	
	$\frac{1}{\sqrt{[\omega_h C_i (r_s \parallel Z_i)]^2 + 1}} = \frac{1}{\sqrt{2}}$	
	$\sqrt{[\omega_h C_i (r_s \parallel Z_i)]^2 + 1} = \sqrt{2}$	
	$\omega_h C_i (r_s \parallel Z_i) = 1$	

$$\omega_H = \frac{1}{C_i(r_s \parallel Z_i)} \text{ rad/s}$$

$$f_H = \frac{1}{2\pi C_i(r_s \parallel Z_i)} \text{ Hz} \quad (10-1)$$

Gain in terms of f_h

$$|A_v| = \frac{A_{v(mid)}}{\sqrt{\left[\frac{\omega}{1/C_i(r_s \parallel Z_i)}\right]^2 + 1}}$$

$$|A_v| = \frac{A_{v(mid)}}{\sqrt{(\omega/\omega_H)^2 + 1}}$$

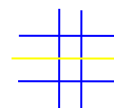
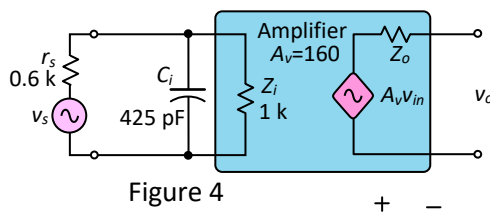
$$|A_v| = \frac{A_{v(mid)}}{\sqrt{(f/f_H)^2 + 1}} \quad (10-2)$$

The shunt capacitance will produce a lagging phase shift from v_s . If we consider $\angle v_s = 0^\circ$, then the angle of v_o (and hence A_v) will be,

$$\angle A_v = -\tan^{-1}(f/f_H) \quad (10-3)$$

Example 10-1

The open circuit voltage gain of the amplifier of Fig.4- is $v_{in}/v_o = 160$. Calculate the overall mid-band voltage gain and the higher cut-off frequency.



Using Equ.(10-)	$A_{v(mid)} = \frac{1000 \times 160 \Omega}{(600 + 1000) \Omega} = 100 \text{ [Ans.]}$	
or,	$A_{v(mid)} = 20 \log_{10} 100 = 40 \text{ dB [Ans.]}$	
Here,	$r_s \parallel r_{in} = \frac{r_s Z_i}{r_s + Z_i} = \frac{600 \times 1000}{600 + 1000} \Omega = 375 \Omega$	
Using Equ.(10-)	$f_H = \frac{1}{2\pi(425 \times 10^{-12}) \times 375} \text{ Hz} \approx 1 \text{ MHz [Ans.]}$	

Comments: If a load is connected at the output, the gain will decrease further.

The high frequency response of the amplifier of Example are plotted in Fig.10-. Fig.10-(a) shows the variation of voltage gain in dB with frequency in a semi-log graph, and Fig.10-(b) shows the variation of phase angel of v_{in} with respect to v_s in another semi-log graph. The mid-band voltage gain ($A_{v(mid)}$) and the higher cut-off frequency (f_H) of the amplifier is 100 (or 40 dB) and 1.7 Hz, respectively. As shown in Fig.10-(a), at the cut-off frequency, the gain is 37 dB, that is 3 dB below the maximum gain. After this cut-off frequency, the gain decreases at a rate of 20 dB/decade or 6 dB/octave. **One octave above the cut-off frequency the gain is 89.5% of mid-band gain, and one octave below it is 44.7%. Again, one decade above f_L the gain is 99.5% of mid-band gain, and one decade below it is 9.95%.** That is, the high frequency response is exactly same as the low frequency response. Only the difference is, here gain decreases with frequency. Fig.10-(b) shows that the phase shift is 45° at the cut-off frequency. One octave above the cut-off frequency the phase shift is 26.6° , and one octave below it is 63.4° . Again, one decade above the cut-off frequency the phase shift is 5.7° , and one decade below it is 84.3° . As shown in Fig.10-(a), the asymptotic response breaks at the point $f = f_H$. For this reason f_H is also called break point frequency. The slope of the asymptote is 6 dB/octave or 20 dB/decade.

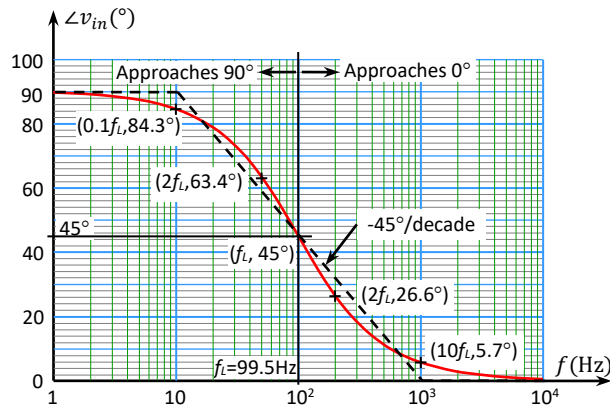


Figure 5

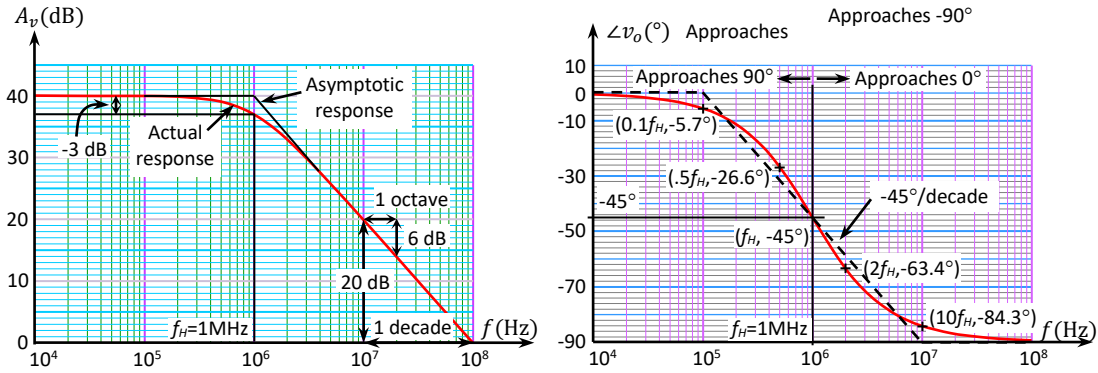


Figure 6

If we consider that the amplifier has no capacitance in the input side and $r_s = 0$, then the response of the output capacitance will be found. Here, in the mid-band frequency range, C_o will work as an open circuit, so

	$v_o = \frac{A_v v_{in} R_L}{Z_o + R_L} \quad \text{or,} \quad \frac{v_o}{v_{in}} = \frac{R_L A_v}{Z_o + R_L}$	
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And

	$\frac{v_{in}}{v_s} = 1$	
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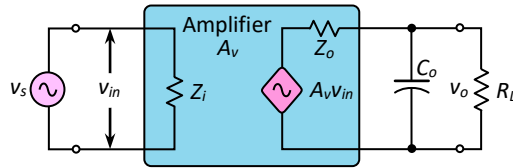


Figure 7

Therefore, the overall mid-band voltage gain will be,

	$A_{v(mid)} = \frac{v_o}{v_s} = \frac{v_{in}}{v_s} \times \frac{v_o}{v_{in}} = 1 \times \frac{Z_i A_v}{r_s + Z_i}$	
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Here also, using the same analysis of the input shunt capacitance, we can show that, the higher cut-off frequency is,

	$\omega_H = \frac{1}{C_o(Z_o \parallel R_L)} \text{ rad/s}$	
or,	$f_{H(C_o)} = \frac{1}{2\pi C_o(Z_o \parallel R_L)} \text{ Hz}$	(10-4)

In terms of the cut-off frequency, the gain will be

	$ A_v = \frac{A_{v(mid)}}{\sqrt{(f/f_{H(C_o)})^2 + 1}}$	(10-5)
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This output shunt capacitance will also produce a lagging phase shift from v_s . If we consider $\angle v_s = 0^\circ$, then the angle of v_o (and hence A_v) will be,

	$\angle A_v = -\tan^{-1}(f/f_H)$	(10-6)
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In any amplifier the input and output shunt capacitances work simultaneously. The high frequency response due to both the capacitances will be same as the two coupling capacitors discussed earlier. If the cut-off frequencies are far apart (at least one decade), the response will have two break points. The overall higher cut-off frequency will be the smaller one. If the two cut-off frequencies are equal, the overall higher cut-off frequency will be 1.55 times the either one. The situation is explained further in Example 10-.

Example 10-2

For the amplifier of Fig.10- calculate the overall mid-band gain, cut-off frequencies due the input and output shunt capacitances and plot the frequency response graph.

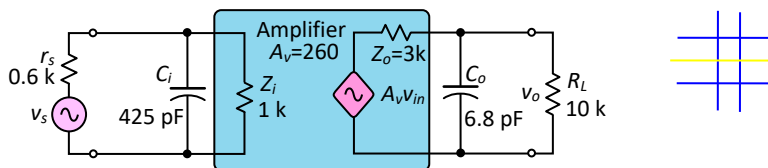


Figure 8

Solution:

The overall mid-band gain of the amplifier will be,

	$A_{v(mid)} = A_v \times \frac{Z_i}{r_s + Z_i} \times \frac{R_L}{Z_o + R_L}$	
or,	$A_{v(mid)} = 260 \times \frac{1000 \Omega}{(600 + 1000) \Omega} \times \frac{10 \text{ k}}{(3 + 10) \text{ k}}$	
$\therefore$	$A_{v(mid)} = 125 \approx 42 \text{ dB [Ans.]}$	

Higher cut-off frequency due to C_i ,

Using Equ.(10-)	$f_H = \frac{1}{2\pi(425 \times 10^{-12}) \times 375} \text{ Hz} \approx 1 \text{ MHz [Ans.]}$	
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Higher cut-off frequency due to C_o ,

Using Equ.	$f_H = \frac{1}{2\pi(6.8 \times 10^{-12})(3k \parallel 10k)} \approx 10 \text{ MHz [Ans.]}$	
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The high frequency response of this amplifier can be drawn using the following expression,

	$ A_v = A_{v(mid)} \times \frac{1}{\sqrt{(f/f_{H(C_i)})^2 + 1}} \times \frac{1}{\sqrt{(f/f_{H(C_o)})^2 + 1}}$	
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The graph of this equation is shown in Fig.10-

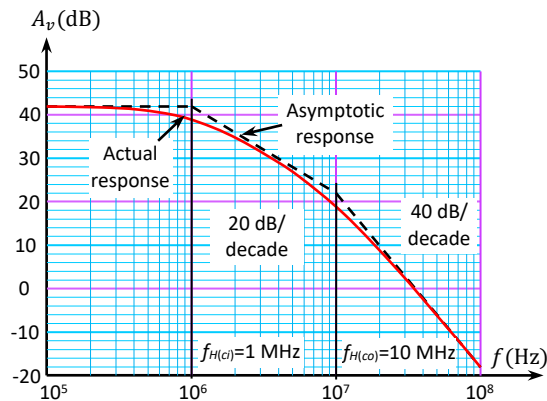


Figure 9

Bandwidth and Gain-Bandwidth Product

The range of frequencies between the lower and higher cut-off frequencies is called *bandwidth*(BW). This is the range of frequencies that the amplifier can properly amplify. Mathematically,

	$BW = f_H - f_L$	(10-7)
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In many amplifiers, like audio amplifiers, the lower cut-off frequency is very small compared to the higher cut-off frequency. In that case, the higher cut-off frequency will be the bandwidth of the amplifier. It is shown in Fig.

The product of gain (generally voltage gain) and the bandwidth of an amplifier is called the *gain-bandwidth product*. It is a Figure of Merit of an amplifier, because it provides the information about the gain of the amplifier and its operating frequency range. For a particular amplifier the gain-bandwidth product is constant. So, if we increase the gain of the amplifier the bandwidth will decrease and vice versa.

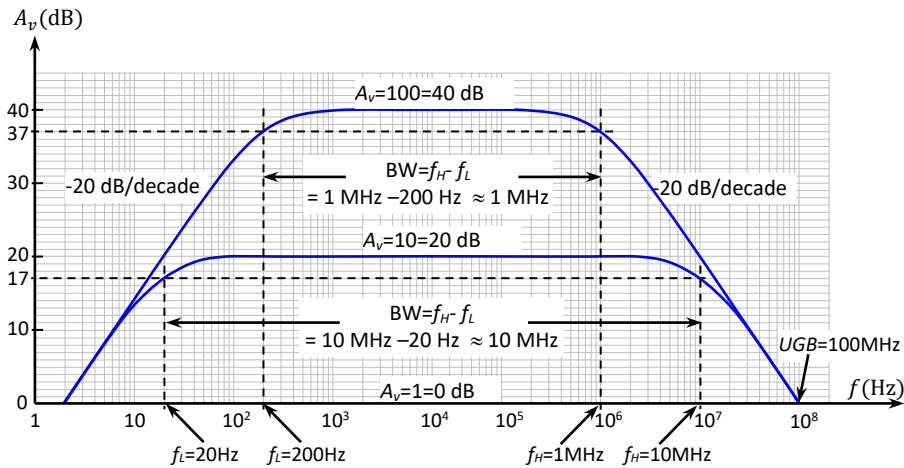


Figure 10

Fig.10- shows the frequency response of an amplifier with a voltage gain $A_v = 100$ or 40 dB, lower cut-off frequency $f_L = 200$ Hz, and higher cut-off frequency $f_H = 1$ MHz. Therefore, the bandwidth of the amplifier will be,

	$BW = f_H - f_L = 1 \text{ MHz} - 200 \text{ Hz} \approx 1 \text{ MHz}$	
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and the gain-bandwidth product will be,

	$GBP = A_v \times BW = 1 \text{ MHz} \times 100 = 100 \text{ MHz}$	
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On the other hand, if the gain of the amplifier is decreased to $A_v = 10$ or 20 dB, the bandwidth will be approximate 10 MHz as shown in the same figure. Therefore, the new GBP will be $10 \text{ MHz} \times 10 = 100 \text{ MHz}$, same as before. Moreover, if the gain of the amplifier is set to *unity*, the gain-bandwidth product will 100 MHz which is called *unity gain bandwidth* (UGB). Note the response of the amplifier: it has only one break point on the low frequency and high frequency side. For amplifiers having only one break points, the gain-bandwidth product is always constant, as found here.

10.1 Frequency Response of Common Emitter Amplifier

The circuit diagram of a common emitter amplifier is given in Fig.10-. Using the hybrid model of transistor the AC equivalent circuit of this amplifier can be drawn as in Fig.10-. In this circuit the inter-electrode capacitances are also shown.

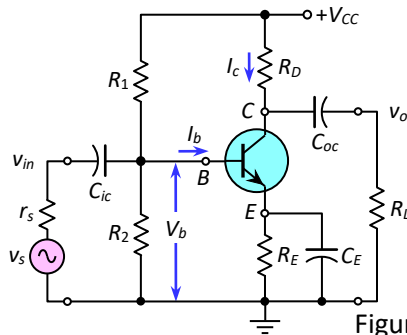


Figure 11

Low Frequency Response

The analysis of this amplifier has been covered in Chapter 5. In that analysis we have calculated the voltage gain, input impedance, output impedance, and current gain. The values are given below.

From Equ.(5-54),	$A_v \approx -\frac{h_{fe}R'_L}{h_{ie}} = -\frac{h_{fe}}{h_{ie}} \left(\frac{R_C R_L}{R_C + R_L} \right)$	
From Equ.(5-57),	$Z_i = R_B \parallel Z_b = R_B \parallel h_{ie}$	
From Equ.(5-59),	$Z_o = r_o \parallel R_C \approx R_C$	

As discussed in Section , the lower cut-off frequency of this amplifier due to the input coupling capacitor will be,

	$f_{L(C_{ic})} = \frac{1}{2\pi(r_s + Z_i)C_{ic}}$	(10-8)
$\therefore$	$f_{L(C_{ic})} = \frac{1}{2\pi(r_s + R_B \parallel h_{ie})C_{ic}} \text{ Hz}$	(10-9)

The lower cut-off frequency of due to the output coupling capacitor will be,

	$f_{L(C_{oc})} = \frac{1}{2\pi(Z_o + R_L)C_{oc}}$	(10-10)
$\therefore$	$f_{L(C_{oc})} = \frac{1}{2\pi(r_o \parallel R_C + R_L)C_{oc}} \text{ Hz}$	(10-11)

The amplifier that uses bypass capacitor will have another lower cut-off frequency due to this bypass capacitor. To calculate this cut-off frequency, we have to calculate the resistance seen by C_E capacitor at point A. $r_s \parallel R_B$ will come in series with the base-to-emitter resistance (h_{ie}). That is total resistance in the base side will be $r_s \parallel R_B + h_{ie}$. When this resistance is seen from

the emitter side it will be divided by h_{fe} and come in parallel with R_E . Therefore, the of total resistance seen at point A will be,

	$R_e = \left(\frac{r_s \ R_B + h_{ie}}{h_{fe}} \right) \ R_E$	
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Where, $R_B = R_1 \| R_2$. The cut-off frequency will occur when the reactance of C_E will be equal to R_e . That is,

	$X_{C_E} = R_e = \left(\frac{r_s \ R_B + h_{ie}}{h_{fe}} \right) \ R_E \text{ [at } f = f_{L(C_E)}]$	
or,	$\frac{1}{2\pi f_{L(C_E)} C_E} = R_e$	
or,	$f_{L(C_E)} = \frac{1}{2\pi R_e C_E}$	
$\therefore$	$f_{L(C_E)} = \frac{1}{2\pi [(r_s \ R_B + h_{ie}) / h_{fe} \ R_E] C_E}$	Put this equ. no in chpa 11

Among the three cut-off frequencies, the overall lower cut-off frequency (f_L) of the amplifier has to be determined as described in Section . If the frequencies are far apart, or one frequency is much greater than the other two, f_L will be the largest of the three. **If two of them are equal and they are larger than the third** one, f_L will be 1.55 times either one. If all are equal, then f_L will be double on any value. However, generally in CE amplifier the cut-off frequency is set by the bypass capacitor C_E .

Example 10-3

Determine the lower and higher cut-off frequencies of this amplifier.

Solution:

To determine r_e we need the value of I_E . To calculate I_E let use check the condition $\beta R_E \gg 10R_2$.

$\beta R_E = 50 \times 2.4 \text{ k} = 120 \text{ k}$, and $10R_2 = 10 \times 27 \text{ k} = 270 \text{ k}$. The condition is not satisfied. So, we have to use exact analysis.

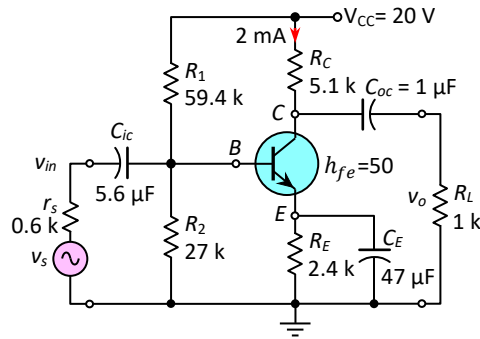


Figure 12

	$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{59.4 \text{ k} \times 27 \text{ k}}{59.4 \text{ k} + 27 \text{ k}} \approx 18.56 \text{ k} = R_B$	(10-12)
	$E_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{20 \text{ V} \times 27 \text{ k}}{59.4 \text{ k} + 27 \text{ k}} \approx 6.25 \text{ V}$	
Using Equ.()	$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (h_{fe} + 1)R_E} = \frac{6.25 \text{ V} - 0.7 \text{ V}}{18.56 \text{ k} + (50 + 1)2.4 \text{ k}} \approx 39.373 \text{ } \mu\text{A}$	

Collector current	$I_C = h_{fe} I_B = 39.373 \text{ } \mu\text{A} \times 50 = 1.97 \text{ mA} \approx I_E$	
$\therefore$	$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.97 \text{ mA}} = 13.2 \text{ } \Omega$	
$\therefore$	$h_{ie} = h_{fe} r_e = 50 \times 13.2 \text{ } \Omega = 0.66 \text{ k}$	
Using Equ.()	$f_{L(C_{ic})} = \frac{1}{2\pi(0.6 \text{ k} + 18.56 \text{ k} 0.66 \text{ k}) \times 5.6 \text{ } \mu\text{F}} \approx 23 \text{ Hz}$	(10-13)
Using Equ.()	$f_{L(C_{oc})} = \frac{1}{2\pi(200 \text{ k} 5.1 \text{ k} + 1 \text{ k}) \times 1 \text{ } \mu\text{F}} \approx 32 \text{ Hz}$	
Using Equ.()	$f_{L(C_E)} = \frac{1}{2\pi[(0.6 \text{ k} 18.56 \text{ k} + 0.66 \text{ k})50 2.4 \text{ k}] \times 47 \text{ } \mu\text{F}} \approx 137 \text{ Hz}$	

As, 137 Hz is much (≈ 1 decade) larger than the other two, the lower cut-off frequency of the amplifier will be $f_L = 137 \text{ Hz}$.

Comments: In Chapter 9, we have found that for CE amplifier the bypass capacitor (C_E) determines the lower cut-off frequency.

High Frequency Response

By using the hybrid- π model of transistor, we can determine the high frequency response of amplifiers. The hybrid- π models of BJT have been discussed in Chapter 5. Using that hybrid- π model we can draw the equivalent circuit of CE amplifier as shown in Fig.

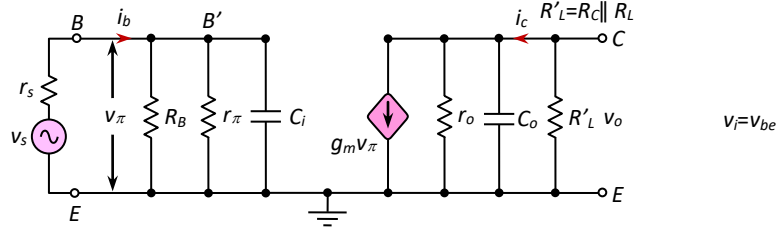


Figure 13

Here, C_i is the total high frequency capacitance in the input side. It includes the base emitter junction capacitance (C_π), Miller capacitance (C_{Mi}) and the wiring capacitance of the circuit (C_{wi}). The voltage gain of the amplifier is $-g_m r_o || R'_L$ and $R'_L = R_C || R_L$. Therefore, the value of the Miller capacitance will be $C_{Mi} = C_\mu (1 + g_m r_o || R'_L)$. That is,

	$C_i = C_\pi + C_{Mi} + C_{wi} = C_\pi + C_\mu (1 + g_m r_o R'_L) + C_{wi}$	(10-14)
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Similarly, C_o is the total high frequency capacitance in output side. It includes the collector emitter junction capacitance (C_{ce}), Miller capacitance (C_{Mo}) and the wiring capacitance of the circuit (C_{wo}). Value of the Miller capacitance will be $C_{Mo} = C_\mu (1 + g_m r_o || R'_L) / (g_m r_o || R'_L) \approx C_\mu$. That is,

	$C_o = C_{ce} + C_{Mo} + C_{wo} = C_{ce} + C_\mu + C_{wo}$	(10-15)
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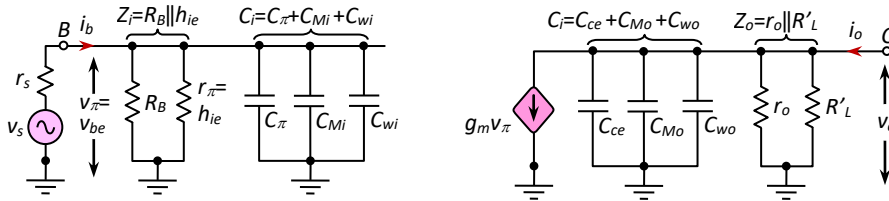


Figure 14

We have already derived the higher cut-off frequencies for the input and output capacitance in topic 10-. The higher cut-off frequency due to the input capacitance is given by Equ.(.),

	$f_{H(C_i)} = \frac{1}{2\pi C_i (r_s Z_i)} \text{ Hz}$	
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Here, the value of input impedance will be,

	$Z_i = R_B r_\pi \approx R_B h_{ie}$	
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$\therefore$	$f_{H(C_i)} = \frac{1}{2\pi C_i(r_s \parallel R_B \parallel h_{ie})} \text{ Hz}$	(10-16)
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If R_B is very large,	$f_{H(C_i)} = \frac{1}{2\pi C_i(r_s \parallel h_{ie})} \text{ Hz} = \frac{r_s + h_{ie}}{2\pi C_i r_s h_{ie}} \text{ Hz}$	(10-17)
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Similarly the higher cut-off frequency due to the output capacitance is given by Equ.(),

	$f_{H(C_o)} = \frac{1}{2\pi C_o(Z_o \parallel R_L)} \text{ Hz}$	
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Here, the value of output impedance will be,

	$Z_o = r_o \parallel R'_L = r_o \parallel R_C \parallel R_L$	
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$\therefore$	$f_{H(C_o)} = \frac{1}{2\pi C_o(r_o \parallel R_C \parallel R_L)} \text{ Hz}$	(10-18)
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As, $r_o \gg R_C \parallel R_L$	$f_{H(C_o)} = \frac{1}{2\pi C_o(R_C \parallel R_L)} \text{ Hz} = \frac{R_C + R_L}{2\pi C_o(R_C R_L)} \text{ Hz}$	(10-19)
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The value of $f_{H(C_i)}$ and $f_{H(C_o)}$ will produce to break points on the asymptote at the high frequency region like the response of Fig. Here also, the overall higher cut-off frequency (f_H) of the amplifier has to be determined as described in Section . If the frequencies (break points) are far apart, then f_H will be the smaller one. If they are equal, then f_H will be 1.55 times either one.

The cut-off frequencies $f_{H(C_i)}$ and $f_{H(C_o)}$ are produced due to the parasitic capacitances of the device and circuit. However, they are not the only parameters that limit the high frequency response of the amplifiers. The value of h_{fe} (or β) of a transistor decreases with frequency in the high frequency region. Therefore, in addition to $f_{H(C_i)}$ and $f_{H(C_o)}$, we must consider the high frequency limitation imposed by the decrease of h_{fe} with frequency. In fact, h_{fe} will produce another higher cut-off frequency (discussed below). Generally, this cut-off frequency would be very larger than $f_{H(C_i)}$ and $f_{H(C_o)}$. If this cut-off frequency is comparable to $f_{H(C_i)}$ and $f_{H(C_o)}$, then the overall higher cut-off frequency of the amplifier has to be determined considering all of the three cut-off frequencies.

But generally for CE amplifier $f_{H(C_i)}$ becomes the higher cut-off frequency of the amplifier, because the value of C_i is larger than C_o due to the Miller effect.

As the value of lower cut-off frequency of an amplifier is very small compared to the higher cut-off frequency, we can consider the higher cut-off frequency as the bandwidth of the amplifier. Therefore, the bandwidth of the CE amplifier will be,

$\therefore$	$BW = \omega_{H(C_i)} = \frac{1}{C_i(r_s \parallel R_B \parallel h_{ie})} \text{ Hz} \approx \frac{r_s + h_{ie}}{C_i r_s h_{ie}} \text{ Hz}$	(10-20)
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and the gain-bandwidth product will be,

$\therefore$	$GBP = A_v \times BW = -\frac{g_m h_{ie} R'_L}{r_s + h_{ie}} \times \frac{r_s + h_{ie}}{C_i r_s h_{ie}} = -\frac{g_m R'_L}{C_i r_s}$	(10-21)
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Example 10-3

Determine the upper cut-off frequency of the amplifier of Example 10-3. Assume that the transistor used in the amplifier has $C_{be} = 10$ pF, $C_{bc} = 5$ pF, and $C_{ce} = 2$ pF. $r_o = 200$ k. neglect wiring capacitances.

Solution:

Here, $C_{be} = C_\pi = 10$ pF, $C_{bc} = C_\mu = 5$ pF and in Example 10-3, we have found $r_e = 13.2 \Omega$

We know that,

	$g_m \approx \frac{1}{r_e} = \frac{1}{13.2 \Omega} \approx 75.76 \text{ mS}$	
--	--	--

Voltage gain of the amplifier will be,

	$A_v = g_m r_o \parallel R_C \parallel R_L = 75.76 \text{ mS} \times (200 \text{ k} \parallel 5.1 \text{ k} \parallel 1 \text{ k}) \approx 63$	
Using Equ.(10-14)	$C_i = 10 \text{ pF} + 5 \text{ pF}(1 + 63) + 0 = 333 \text{ pF}$	
Using Equ.(10-15)	$C_o = C_{ce} + C_{M_o} + C_{w_o} = 2 \text{ pF} + 5 \text{ pF} + 0 = 7 \text{ pF}$	
Using Equ.(10-16)	$f_{H(C_i)} = \frac{1}{2\pi \times 333 \text{ pF}(0.6 \text{ k} \parallel 18.56 \text{ k} \parallel 0.66 \text{ k})} \approx 1.55 \text{ MHz}$	
Using Equ.(10-18)	$f_{H(C_o)} = \frac{1}{2\pi \times 7 \text{ pF}(200 \text{ k} \parallel 5.1 \text{ k} \parallel 1 \text{ k})} \text{ Hz} \approx 27 \text{ MHz}$	

Between these two frequencies, the effective higher cut-off frequency will be the smaller one, that is $f_H = 1.5$ MHz.

Comments: In CE amplifier generally, the input parasitic capacitance determine the higher cut-off frequency, due to the Miller's effect.

Broadband or Wideband Amplifier

Broadband amplifiers are also called wideband amplifiers. These amplifier are designed to provide flat gain over a wide range of frequencies, often from DC to 10 MHz and perhaps 200 MHz or more with voltage gain of about 100 or 40 dB. These amplifiers are used to amplify

video signal and their large bandwidth permits faithful reproduction of exact input and output. Broadband amplifiers which are also called video amplifiers find extensive use in TV, RADAR and many other instrument circuits.

High Frequency Compensation of CE Amplifiers

The frequency of single stage CE amplifiers or capacitor-coupled multistage amplifiers can be increased by compensating the internal capacitance effects of the transistors. The most common compensation techniques are:

- i) High frequency base compensation
- ii) High frequency shunt compensation
- iii) High frequency series compensation

High Frequency Base Compensation:

In this technique a parallel connected capacitor and resistor is used at the base of the amplifier. Fig.15 shows a base compensated CE amplifier. Here all bypass and coupling capacitors are considered as short circuit. The high frequency equivalent circuit of this amplifier is shown in Fig.16. Considering $R_B > h_{ie}$, it has been neglected.

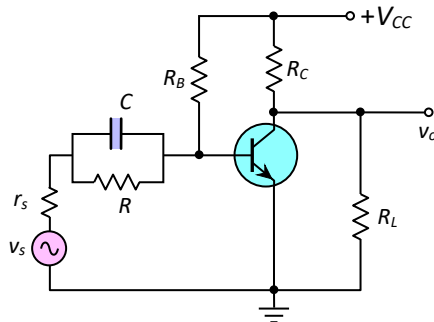


Figure 15

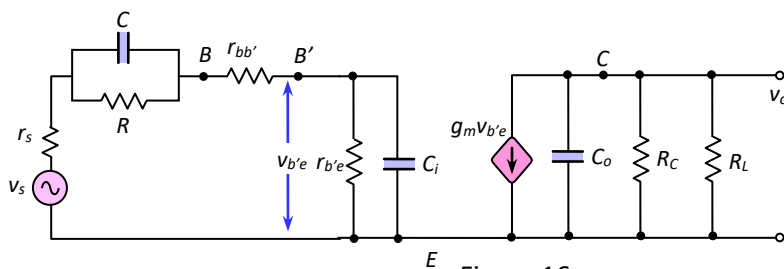


Figure 16

At mid-band frequencies, all the capacitors will be open. So, the mid-band gain of the amplifier will be,

	$A_{v(mid)} = -\frac{r_{b'e}g_m R_{L'}}{r_s + R + r_{bb'} + r_{b'e}}, \quad \text{where } R_{L'} = R_C \parallel R_L$	(10-22)
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Now we will determine the higher cut-off frequency. Let us assume Z and $Z_{b'e}$ be the equivalent impedance of the parallel RC and the impedance across $B'E$.

$\therefore$	$Z = R \parallel C = \frac{R(1/j\omega C)}{R + (1/j\omega C)}$	
<i>or,</i>	$Z = \frac{R}{1 + j\omega RC}$	(10-23)
Similarly,	$Z_{b'e} = \frac{r_{b'e}}{1 + j\omega r_{b'e} C_i}$	

For proper compensation the time constants should be same, that is, $r_{b'e}C_i = RC$.

$\therefore$	$Z_{b'e} = \frac{r_{b'e}}{1 + j\omega RC}$	(10-24)
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As $C_i \gg C_o$, the higher cut-off frequency will be imposed by C_i . Considering $C_o = 0$, gain at high frequency will be,

	$A_{v(HF)} = -\frac{Z_{b'e}g_m R_{L'}}{r_s + Z + r_{bb'} + Z_{b'e}}$	(10-25)
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Putting the value of Z and $Z_{b'e}$

	$A_{v(HF)} = -\frac{(r_{b'e}/1 + j\omega RC)g_m R_{L'}}{r_s + (R/1 + j\omega RC) + r_{bb'} + (r_{b'e}/1 + j\omega RC)}$	
<i>or,</i>	$A_{v(HF)} = -\frac{r_{b'e}g_m R_{L'}}{r_s(1 + j\omega RC) + R + r_{bb'}(1 + j\omega RC) + r_{b'e}}$	
<i>or,</i>	$A_{v(HF)} = -\frac{r_{b'e}g_m R_{L'}}{r_s + j\omega r_s RC + R + r_{bb'} + j\omega r_{bb'} RC + r_{b'e}}$	
<i>or,</i>	$A_{v(HF)} = -\frac{r_{b'e}g_m R_{L'}}{(r_s + R + r_{bb'} + r_{b'e}) + j\omega RC(r_s + r_{bb'})}$	
<i>or,</i>	$A_{v(HF)} = -\frac{r_{b'e}g_m R_{L'}/(r_s + R + r_{bb'} + r_{b'e})}{1 + j\omega RC(r_s + r_{bb'})/(r_s + R + r_{bb'} + r_{b'e})}$	
<i>or,</i>	$A_{v(HF)} = \frac{A_{v(mid)}}{1 + j\omega/\omega_H}$	
where,	$\omega_H = (r_s + R + r_{bb'} + r_{b'e})/RC(r_s + r_{bb'})$	
<i>or,</i>	$\omega_H = (r_s + R + r_{bb'} + r_{b'e})/r_{b'e}C_i(r_s + r_{bb'})$	(10-26)

Magnitude of $A_{v(HF)}$ will be,

	$ A_{v(HF)} = \frac{A_{v(mid)}}{\sqrt{1 + (\omega/\omega_H)^2}}$	(10-27)
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Therefore, when $\omega = \omega_H$, then

	$ A_{v(HF)} = \frac{A_{v(mid)}}{\sqrt{1 + (1)^2}} = \frac{A_{v(mid)}}{\sqrt{2}}$	(10-28)
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So, ω_H is the higher cut-off frequency of the compensated amplifier which is given by Equ.(.). From this equation, we find that ω_H increases with the increase in R . Bandwidth of the compensated amplifier will be

	$BW \approx \omega_H = \frac{r_s + R + r_{bb'} + r_{b'e}}{r_{b'e}C_i(r_s + r_{bb'})}$	(10-29)
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Neglecting $r_{bb'}$ and using $r_{b'e} \approx h_{ie}$, we can also write this expression as,

	$BW = \frac{r_s + R + h_{ie}}{r_s h_{ie} C_i}$	(10-30)
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Gain bandwidth product will be,

	$GBP = A_{v(HF)} \times BW$	(10-31)
--	-----------------------------	---------

	$GBP = - \frac{r_{b'e} g_m R_{L'}}{r_s + R + r_{bb'} + r_{b'e}} \frac{r_s + R + r_{bb'} + r_{b'e}}{r_{b'e} C_i (r_s + r_{bb'})}$	(10-32)
--	--	---------

	$GBP = - \frac{g_m R_{L'}}{C_i (r_s + r_{bb'})}$	(10-33)
--	--	---------

Neglecting $r_{bb'}$, we can also write this expression as,

	$GBP = - \frac{g_m R_{L'}}{C_i r_s}$	(10-34)
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Compare with uncompensated BW and GBP

	$BW = \frac{r_s + h_{ie}}{C_i r_s h_{ie}} \text{ Hz}$	
	$GBP = - \frac{g_m R_{L'}}{C_i r_s}$	

Therefore, we can conclude that the GBP remains same but the BW increases.

10.2 High Frequency Compensation using Shunt Peaking

We have already discussed that in a single stage CE amplifier the value of output parasitic capacitance (C_i) is lower than that of input parasitic capacitance (C_o). Therefore, we have neglected the effect of C_i . But in a multi-stage amplifier, the input capacitance of the second stage is added with the output capacitance of the first stage. Then the total output capacitance becomes even higher than the input capacitance and reduces the bandwidth. One method to eliminate the effect of this output capacitance is to use a small inductance in series with the load resistor (R_C). As the added inductance comes in parallel with the output capacitance this method is called *high frequency shunt compensation*. Fig.17 shows a two-stage CE amplifier with shunt compensation. Here, C_{o1} is the output capacitance of the first stage and C_{i2} is the input capacitance of the second stage and are given by Equ.(10-14), and Equ.(10-15) respectively.

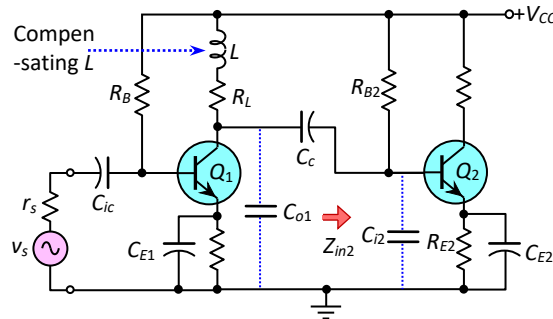


Fig.17: Two-stage CE amplifier compensated with shunt peaking

The added inductance works in parallel with the output capacitance. The value of L is so chosen that it resonates with output capacitance at near the cut-off frequency of the uncompensated amplifier. Thus the overall impedance increases and so does the gain of the amplifier.

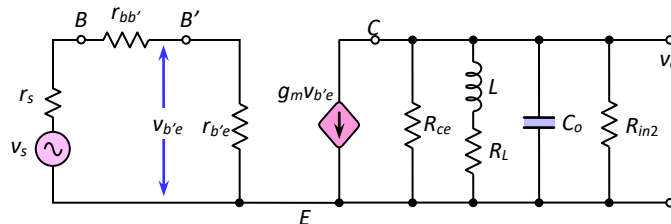


Figure 18

The high frequency equivalent circuit is shown in Fig.18. Here, C_o is the total capacitance effective at the output of the first stage, that is, $C_o = C_{o1} + C_{i2}$ and R_{in2} is the resistance of the second stage. The value of R_B has been considered very large.

Generally R_{ce} is very large. We will assume the input impedance of the second stage, R_{in2} is also very large. Neglecting these two resistors, we can further simplify the circuit as in Fig.19.

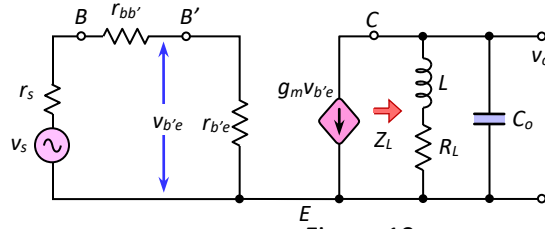


Figure 19

The effective load at the collector of the circuit will be,

	$Z_L = (R_L + j\omega L) \parallel (1/j\omega C_o)$	
or,	$Z_L = \frac{(R_L + j\omega L)(1/j\omega C_o)}{(R_L + j\omega L) + (1/j\omega C_o)}$	
Multiplying by $j\omega C_o$,	$Z_L = \frac{R_L + j\omega L}{(j\omega C_o R_L - \omega^2 L C_o) + 1}$	
Multiplying by $R_L - j\omega L$,	$Z_L = \frac{R_L^2 + \omega^2 L^2}{j\omega C_o R_L (R_L - j\omega L) - \omega^2 L C_o (R_L - j\omega L) + R_L - j\omega L}$	
or,	$Z_L = \frac{R_L^2 + \omega^2 L^2}{j\omega C_o R_L^2 + \omega^2 C_o R_L L - \omega^2 L C_o R_L + j\omega^3 L^2 C_o + R_L - j\omega L}$	
or,	$Z_L = \frac{R_L^2 + \omega^2 L^2}{j\omega [C_o (R_L^2 + \omega^2 L^2) - L] + R_L}$	(10-35)

To make the circuit independent of reactance,

	$j\omega [C_o (R_L^2 + \omega^2 L^2) - L] = 0$	
	$C_o (R_L^2 + \omega^2 L^2) = L$	
	$C_o = \frac{L}{R_L^2 + \omega^2 L^2}$	

Generally the added inductance is very small compared to R_L . Therefore, $R_L^2 \gg \omega^2 L^2$

$\therefore$	$C_o = \frac{L}{R_L^2}$	
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$\therefore$	$L = R_L^2 C_o$	
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Therefore, from Equ.(.),

$\therefore$	$Z_L = \frac{R_L^2 + \omega^2 L^2}{0 + R_L}$	
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At the mid-frequency range C_o will be open and L is almost short. Therefore, from Equ.(10-35) we get,

$\therefore$	$Z_L = \frac{R_L^2 + \omega^2 L^2}{0 + R_L}$	
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Therefore, the output voltage will be

$\therefore$	$v_o = -g_m v_{b'e} Z_L = -g_m v_{b'e} \frac{R_L^2 + \omega^2 L^2}{R_L}$	
$\therefore$	$\frac{v_o}{v_{b'e}} = -g_m \frac{R_L^2 + \omega^2 L^2}{R_L}$	

From the input side of Fig.19,

	$v_{b'e} = v_s \frac{r_{b'e}}{r_s + r_{bb'} + r_{b'e}}$	
	$\frac{v_{b'e}}{v_s} = \frac{r_{b'e}}{r_s + r_{bb'} + r_{b'e}}$	

Therefore, the mid-frequency voltage gain of the amplifier will be,

	$A_{v(MF)} = \frac{v_{b'e}}{v_s} \times \frac{v_o}{v_{b'e}} = -g_m \frac{R_L^2 + \omega^2 L^2}{R_L} \frac{r_{b'e}}{r_s + r_{bb'} + r_{b'e}}$	
--	--	--

But as $r_{b'e} \gg r_s + r_{bb'}$, we can neglect $r_s + r_{bb'}$.

	$A_{v(MF)} = -g_m \frac{R_L^2 + \omega^2 L^2}{R_L}$	(10-36)
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Voltage gain at high frequency:

	$A_{v(HF)} = \frac{-g_m(R_L^2 + \omega^2 L^2)}{j\omega[C_o(R_L^2 + \omega^2 L^2) - L] + R_L}$	
$\therefore$	$ A_{v(HF)} = \frac{g_m(R_L^2 + \omega^2 L^2)}{\sqrt{R_L^2 + \omega^2[C_o(R_L^2 + \omega^2 L^2) - L]^2}}$	
or,	$ A_{v(HF)} = \frac{g_m(R_L^2 + \omega^2 L^2)}{\sqrt{R_L^2 + \omega^2 C_o^2(R_L^2 + \omega^2 L^2)^2 - 2\omega^2 L C_o(R_L^2 + \omega^2 L^2) + \omega^2 L^2}}$	

or,	$ A_{v(HF)} = \frac{g_m(R_L^2 + \omega^2 L^2)}{\sqrt{(R_L^2 + \omega^2 L^2)}\sqrt{1 + \omega^2 C_o^2(R_L^2 + \omega^2 L^2) - 2\omega^2 L C_o}}$	
or,	$ A_{v(HF)} = \frac{g_m R_L \sqrt{(1 + \omega^2 L^2 / R_L^2)}}{\sqrt{1 + \omega^2 R_L^2 C_o^2(1 + \omega^2 L^2 / R_L^2) - 2\omega^2 L C_o}}$	
Let,	$L = h R_L^2 C_o$	(10-37)
where, h is a constant	$h = L / R_L^2 C_o$	(10-38)

Putting the value of L

	$ A_{v(HF)} = \frac{g_m R_L \sqrt{(1 + \omega^2 (h R_L^2 C_o)^2 / R_L^2)}}{\sqrt{1 + \omega^2 R_L^2 C_o^2(1 + \omega^2 (h R_L^2 C_o)^2 / R_L^2) - 2\omega^2 (h R_L^2 C_o) C_o}}$	
or,	$ A_{v(HF)} = \frac{g_m R_L \sqrt{(1 + \omega^2 h^2 R_L^2 C_o^2)}}{\sqrt{1 + \omega^2 R_L^2 C_o^2(1 + \omega^2 h^2 R_L^2 C_o^2) - 2\omega^2 h R_L^2 C_o^2}}$	(10-39)

Without compensation the higher cut-off frequency would have been

	$\omega_H = \frac{1}{R_L C_o}$	(10-40)
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Putting the value of ω_H into Equ.(10-39) of $|A_{v(HF)}|$, we get

	$ A_{v(HF)} = \frac{g_m R_L \sqrt{(1 + \omega^2 h^2 / \omega_H^2)}}{\sqrt{1 + \omega^2 / \omega_H^2(1 + \omega^2 h^2 / \omega_H^2) - 2\omega^2 h / \omega_H^2}}$	
$\therefore$	$ A_{v(HF)} = \frac{g_m R_L \sqrt{[1 + (\omega / \omega_H)^2 h^2]}}{\sqrt{1 + (\omega / \omega_H)^2 [1 + (\omega / \omega_H)^2 h^2] - 2h(\omega / \omega_H)^2}}$	
$\therefore$	$ A_{v(HF)} = \frac{g_m R_L \sqrt{[1 + (\omega / \omega_H)^2 h^2]}}{\sqrt{1 + (\omega / \omega_H)^2 (1 - 2h) + h^2(\omega / \omega_H)^4}}$	
$\therefore$	$ A_{v(HF)} = \frac{g_m R_L}{\sqrt{\frac{1 + (\omega / \omega_H)^2 (1 - 2h) + h^2(\omega / \omega_H)^4}{[1 + (\omega / \omega_H)^2 h^2]}}}$	
$\therefore$	$ A_{v(HF)} = \frac{ A_{v(MF)} }{\sqrt{\frac{1 + (\omega / \omega_H)^2 (1 - 2h) + h^2(\omega / \omega_H)^4}{[1 + (\omega / \omega_H)^2 h^2]}}}$	

The compensated higher cut-off frequency (Let ω_{HC}) will be obtained when the denominator the equation is $\sqrt{2}$.

$\therefore$	$\sqrt{\frac{1 + (\omega_{HC}/\omega_H)^2(1 - 2h) + h^2(\omega_{HC}/\omega_H)^4}{[1 + (\omega_{HC}/\omega_H)^2h^2]}} = \sqrt{2}$	
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Let us consider the value of L is so chosen that $h = 0.5$, then from equation

$\therefore$	$\frac{1 + 0.25(\omega_{HC}/\omega_H)^4}{1 + 0.25(\omega_{HC}/\omega_H)^2} = 2$	
$\therefore$	$1 + 0.25(\omega_{HC}/\omega_H)^4 = 2 + 0.5(\omega_{HC}/\omega_H)^2$	
$\therefore$	$1 + 0.5(\omega_{HC}/\omega_H)^2 - 0.25(\omega_{HC}/\omega_H)^4 = 0$	

Solving the above equation, we get,

	$\omega_{HC}/\omega_H \approx 1.8$	
$\therefore$	$\omega_{HC} = 1.8\omega_H$	

This shows that for $h = 0.5$ the compensated higher cut-off frequency has been 1.8 times the uncompensated higher cut-off frequency.

$\therefore$	$ A_{v(HF)} = \frac{ A_{v(MF)} }{\sqrt{\frac{1 + (\omega/\omega_H)^2(1 - 2h) + h^2(\omega/\omega_H)^4}{[1 + (\omega/\omega_H)^2h^2]}}}$	
--------------	--	--

At $\omega = \omega_H$ the gain of the amplifier can be found from Equ. (shown above)

$\therefore$	$ A_{v(HF)} = \frac{ A_{v(MF)} }{\sqrt{\frac{1 + (1 - 2h) + h^2}{1 + h^2}}}$	
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For $h = 0.5$

	$ A_{v(HF)} = A_{v(MF)} = \sqrt{2} A_{v(MF)} /\sqrt{2}$	
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Without compensation at $\omega = \omega_H$ the amplifier's gain would have been $|A_{v(MF)}|/\sqrt{2}$. Now due to compensation, and for $h = 0.5$, the gain of the compensated amplifier has been increased by a factor of $\sqrt{2}$. The compensated higher cut-off frequency and the gain of the amplifier depends on the value of L and hence on the value of h . As shown in Fig.20, the higher the value of h the higher the gain of the amplifier at $\omega = \omega_H$

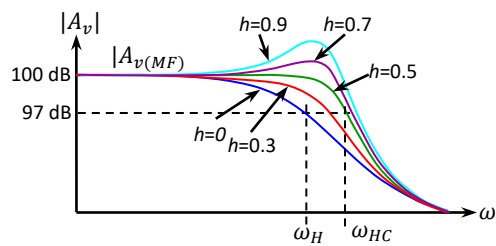


Figure 20